

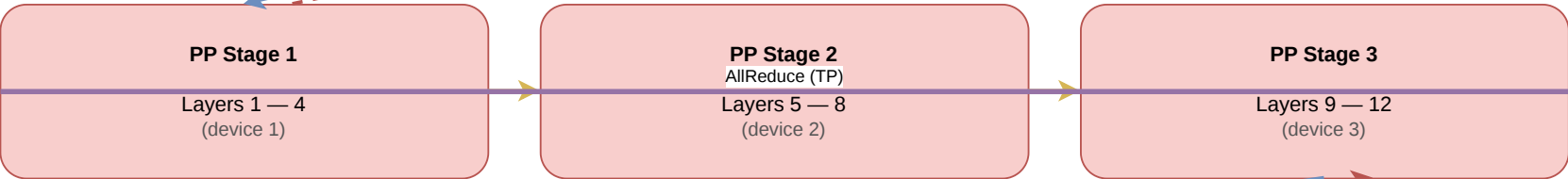
Distributed Transformer Training: DP + TP + PP Combined (background)

Forward + Backward + Parameter Sync across DP / TP / PP

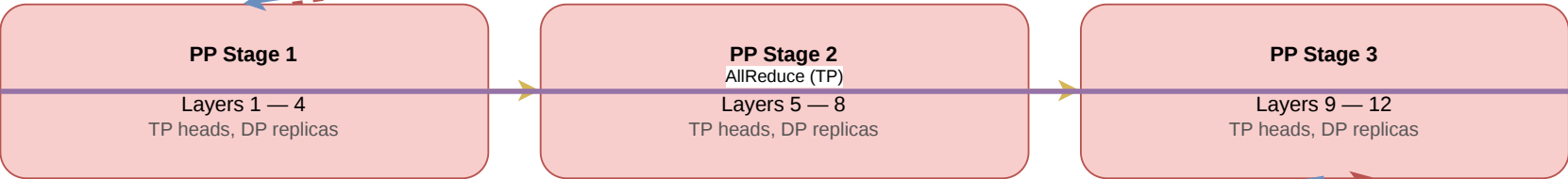
Legend

- forward data flow
- backward gradient
- AllReduce param sync
- pipeline activation

① Encoder — Pipeline Parallel (12 layers split into 3 stages, each stage on a separate device)



② Decoder — Hybrid (PP × TP × DP, attention heads split via TP within each PP stage)



Output Layer

DP + TP (logits projected, vocabulary partitioned)

③ Sharded Optimizer State (ZeRO style)

load weights

checkpoint

Optimizer State Shards

each DP rank holds 1/N of optimizer state (momentum, variance) — ZeRO Stage 3